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¹R: Document, report (excluding the periodic and final reports); DEM: Demonstrator, pilot, prototype, plan designs; DEC: Websites, patents filing, press & media actions, videos, etc.; DATA: Data sets, microdata, etc.; DMP: Data management plan; ETHICS: Deliverables related to ethics issues.; SECURITY: Deliverables related to security issues; OTHER: Software, technical diagram, algorithms, models, etc.

²PU – Public, fully open, e.g. web (Deliverables flagged as public will be automatically published in CORDIS project's page); SEN – Sensitive, limited under the conditions of the Grant Agreement; Classified R-UE/EU-R – EU RESTRICTED under the Commission Decision No2015/444; Classified C-UE/EU-C – EU CONFIDENTIAL under the Commission Decision No2015/444; Classified S-UE/EU-S – EU SECRET under the Commission Decision No2015/444

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Abstract

This document introduces ethical resilience as a guiding concept for human–AI interaction in light of two emerging developments: the transition from singular to collective agentive AI systems, and the increasing relevance of machine awareness as a functional, non-phenomenal descriptor of AI capacities. It argues that these developments raise novel ethical challenges that are not fully addressed by existing AI ethics frameworks, particularly with respect to cooperation, responsibility, communication, and explainability. Building on a distinction between awareness and consciousness, the document examines how collective and aware agentive systems reshape human expectations, trust, and oversight. It concludes by proposing practical guidelines to support ethically resilient human–AI relations, aiming to inform future research, system design, and regulatory approaches for emerging collective and aware artificial agents.

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1. Introduction

Artificial intelligence (AI) research is advancing rapidly, and ethicists, particularly those engaged in policy development, are working to keep pace (Coeckelbergh, 2020; Nyholm, 2022). This paper argues for expanding the ethical debate to address two emerging developments in robotics and AI: the move from singular to collective agentive systems, and the growing relevance of machine awareness as a descriptor of these systems' functional capacities (Figure 1). It highlights key risks and opportunities that arise as the public begins to interact with collective, rather than singular, and functionally aware or partially aware (rather than non-aware) agentive AI systems.

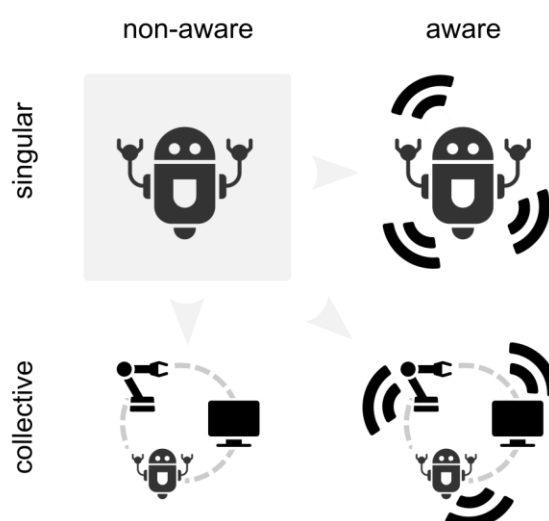


Figure 1 Two avenues of novelty. We consider a shift from singular to collective and from non-aware to aware systems in human interactions with AI. The move from non-aware to aware systems does not necessarily reflect a binary distinction. Awareness may be a graded and multidimensional state or feature of a system. Similarly, the shift from singular to collective systems can be further nuanced. For example, a collective system may consist of several connected individually aware components. Alternatively, collective awareness may be an emergent property of the system as a whole that supervenes on properties of its individual components.

Much of the existing AI ethics literature focuses on human interactions with non-aware automated systems—or systems that have so far been considered non-aware. This is largely because awareness has traditionally been construed as closely tied to consciousness (e.g., Block, 1995; Seth and Bayne, 2022) which most researchers agree that current technology have not achieved (Aru et al., 2023; Butlin et al, 2023; Chalmers, 2023; Schwitzgebel, 2023). That said, it has also been proposed to distinguish the two notions (e.g., Chalmers, 1995) and several large-scale interdisciplinary research projects are currently exploring to revise conceptions of awareness in both humans and machines.³ Importantly, some of these projects

³ A notable example is the suite of research projects funded by the European Innovation Council under the Pathfinder Challenge program titled 'Awareness Inside': <https://awarenessinside.eu/>.

explicitly distinguish machine awareness from machine consciousness and focus specifically on the former (Bacciu et al., 2025; Evers et al., 2025). These developments call for a preemptive ethical examination of human interactions with AI systems that are presented to human stakeholders in terms of novel concepts of awareness.

Also, AI ethics has largely addressed challenges arising in human interactions with individual or singular systems. However, futuristic visions of people relying on networks of personalized, interconnected agentive systems to collectively accomplish tasks—coupled with rapid advances in swarm robotics—call for closer examination of the ethics of human interactions with collective, rather than singular, AI-powered tools.

To motivate new ethical considerations that arise from the two avenues of novelty introduced by the shift from singular to collective and from non-aware to aware systems in human interactions with AI (Figure 1), we discuss examples that focus on four key topics: cooperation, responsibility, communication, and explainability (Figure 2).



Figure 2. Four areas of focus in human interaction with AI after the introduction of the two avenues of novelty in Figure 1.

Some of these concerns are already recognised in existing work on the ethics of human interactions with non-aware, singular AI systems and have been incorporated into regulatory frameworks such as the European Union’s Artificial Intelligence Act. First proposed by the European Commission in 2021 and adopted in its final form in 2024 (Regulation (EU) 2024/1689), the AI Act takes a deliberately human-centred and risk-based approach. It regulates AI systems without attributing awareness, agency, or moral status to them, focusing instead on protecting human users and those indirectly affected by their deployment, with

particular attention to risks of exploitation and harm to vulnerable populations (see also O'Neil, 2016). Our approach is similar in spirit, but it highlights new points, and our treatment of human cooperation with agentic AI systems is novel.

In the remainder of this document, we will address the four topics in turn and provide recommendations for future research on each in the context of collective and/or aware agentic systems. Our analysis of the associated risks and opportunities is by no means exhaustive; rather, our goal is to spearhead a broadening of the ethical discourse to encompass the two avenues of novelty discussed here. First, however, we briefly introduce several conceptual clarifications. We explain what we mean by collective, as opposed to singular, agentic systems; introduce one approach to disentangling awareness from consciousness; and outline several benefits of using machine awareness as a descriptor of agentic systems' functional capacities.

2. Two novelties: From singular to collective and from non-aware to aware systems

2.1 Collective versus singular agentic systems

By an *agentic system*, we mean a decision-making entity that exhibits goal-directed or purposive behaviour, selects actions on the basis of internal states (e.g., representations and objectives), interacts with its environment, other agentic systems, and/or humans in ways that resemble agency, and can operate to varying degrees independently of direct human steering (Russell and Norvig, 2021). We use the term agentic in a functional sense that does not automatically imply moral agency or responsibility. By an agentic AI system, we mean an agentic system that is fully or partly powered by AI, and we use the terms agentic AI system and AI agent interchangeably.

We use the terms *singular*, *unitary*, and *individual* interchangeably to refer to an agentic system that pursues its objective independently, without relying on other systems. In contrast, by a *plural* or *collective agentic system* we mean a system that operates as part of a larger group of systems that collectively pursue a shared objective. For example, an automated (self-driving) taxi might be tasked with safely and efficiently transporting as many passengers as possible within its service area over a given timeframe. The optimal strategies it adopts as an individual unit may differ significantly from those it would adopt as part of a coordinated fleet pursuing this objective jointly. In the latter case, what matters is the total number of passengers transported across the service area, rather than the number transported by any single taxi.

A *collective agentic system* can also be understood as a system whose individual components jointly act—or are treated—as a single agentic entity at the collective level. Construed in this way, agency is attributed at the level of the system as a whole, which pursues its objective through the coordinated actions of its components. Whether the objectives, actions, and other states of such a system can be reduced to those of its individual components is a matter of debate, which we do not address here (see, e.g., Gilbert, 1989; List and Pettit, 2011). While the distinction between *plural* and *collective* agentic systems can be useful, for simplicity we use the term collective agentic system to encompass both of these understandings, and we indicate the relevant distinction where it matters.

2.2 Awareness as distinct from consciousness

Philosophical debate and everyday usage of the terms often treat *awareness* and *consciousness* interchangeably, or see awareness as an aspect of consciousness (Zeman, 2006; Levy, 2011; Seth and Bayne, 2022). Chalmers (1995) challenged this tendency, proposing awareness to be a somewhat simpler notion than consciousness. Awareness, according to him, allows one to explain what he called “easy problems,” for example, an agent’s ability to discriminate, categorise, and react to environmental stimuli. However, in practice, the difficulty in disentangling awareness from consciousness remains due to their ambiguous usage to date (Black, 2017).

Theories of consciousness vary widely, ranging from fairly minimal to highly demanding conceptions of what it means to be conscious. Depending on one’s theoretical commitments, consciousness studies may emphasise anything from the integration of information (Albantakis et al., 2023) to meta-representational abilities such as the capacity to think about one’s own mental states (Rosenthal, 2009). In the phenomenal sense, consciousness broadly refers to the qualitative, subjective character of experience—what it is like to see red, feel pain, or hear music (Nagel, 1974). In contrast, access consciousness (Block, 1995) emphasises the (conscious) availability of information for reasoning, rational speech, and action. These various conceptions and theories differ not only in emphasis, but also in their explanatory goals and underlying assumptions, often leading to widely varying—and sometimes incompatible—criteria for identifying consciousness in natural systems, let alone generating or assessing it in artificial ones.

When it comes to artificial agents in particular, it is worthwhile to assess their perceptual and action-guiding abilities while remaining neutral on the question of whether they possess subjective experience or ‘inner lives’. Some have even argued that we may have ethical reasons to avoid designing artificial agents with the capacity for phenomenal consciousness, in order to prevent the possibility of artificial suffering (e.g., Metzinger, 2021; Schwitzgebel, 2023). This motivates a distinction between *phenomenal consciousness* and *awareness*, but also between *awareness* and *access consciousness*, which ties information availability to conscious access in ways that are both potentially too demanding, and plain unnecessary for artificial agents. This distinction between forms of consciousness and awareness is further supported by recent systematic work on AI awareness (Zhu and Hussain, 2024), which synthesizes research on how the concept of “awareness” is used and understood across disciplines in studies from 2019 to 2025.

Awareness, as understood in recent work (Lee et al., 2025, Lee et al 2026), provides a flexible basis for evaluating action-perception abilities across diverse systems, without presupposing phenomenal states or second-order access. It focuses on an agent’s capacities to perceive and act upon information, measured through observable behaviour in a range of tasks. A dynamic, graded, multidimensional formulation and ability-centred formulation of awareness centred around abilities allows for *fair* systematic comparison across agents—acknowledging the need for species- or system-specific task alignment to accommodate varying constraints (Firestone, 2020). In this light, awareness could reveal the type(s) of (adaptive) strategies available to a system in engaging with environmental events, internal states, or social cues.

This pragmatic account of awareness entails a functional approach, focused on how and to what extent systems interact with others, themselves, and their environment. Take, for instance, a vacuuming robot. We can ask how aware it is of the space it is navigating through, and the type of objects and other agents that it encounters in that space, by assessing how successfully it can adapt to changes (for example, the introduction of a new couch or the presence of another moving agent) without being interested in what, if anything, it might feel like for the robot to be able to do that. This approach largely aligns with Rahwan's et al. (2019) call for establishing a scientific discipline of machine behaviour that employs empirical methods from, for instance, ethology and behavioural ecology, to study AI agents through an integrated, interdisciplinary lens. Beyond this, it is also roughly in line with an approach recently advocated in AI ethics research to assess and determine what status could or should be attributed to machines based on their performance in relevant domains (Danaher, 2019, 2021; see also Nyholm, 2022, chapters 8 and 9).

2.3 Benefits of machine awareness

There are numerous potential benefits from developing and introducing the concept of machine awareness to people's interactions with AI. One of them we already mentioned. Knowing an AI system's awareness dimensions and its degrees of awareness in those dimensions should make it easier for human users of that system to rely on and delegate tasks to it. For example, knowing that one's fully automated (self-driving) vehicle has a high degree of awareness, enabling it to adaptively respond to the variety of traffic participants it may encounter—pedestrians, cyclists, other vehicles, careless children playing ball close to schools, and so on—would make it easier for one to delegate the task of driving to it in busy urban traffic. Conversely, knowing that one's conversational AI assistant has a low degree of awareness, reflected in its limited ability to assess the truthfulness of its boldly stated claims on a particular topic would make one rightly more sceptical about the assistant's communicated "facts."

Relatedly, accurately and clearly communicated awareness dimensions of a machine, and its degrees of awareness in those dimensions, will allow its human users and supervisors to have a better understanding of where human oversight is the most important and required in collaborative human-machine teams. For example, suppose that one's job requires one to speedily disseminate important information to a wide audience in multiple languages and that one does that with the help of an automated translation tool. Knowing gaps in the translation tool's awareness—specifically in its abilities to recognise important nuances in specific languages—helps identify which topics and languages require more careful human oversight and checking (and concerning which topics and languages the tool can be safely relied upon in this collaborative work).

Machine awareness can also safeguard people from unintended and unwelcome side effects of automation. An automated system's awareness, demonstrated through its handling of potential problems associated with factors on which it bases its decisions and which therefore contribute to its performance success, allow it to flag those problematic cases to its human supervisors and thereby help avoid potentially disastrous outcomes (whether in the domain of recommending fitness training plans to human users or in filtering job applicants' CVs in a human resources department). This is particularly relevant in the context of automated nudging of human behaviours (Sætra, 2019; Schmauder et al., 2023). Simply put, the problem

with unaware AI systems is that, without the capacity to monitor, adapt, and adjust the ways in which they achieve their tasked objectives in response to environmental, social, or internal information, they may generate unintended side effects that go unnoticed. Crucially, in order to be able to flag a potential problem with what one does, one needs to be aware of that potential problem in the first place.

3. Four key topics in human–AI interaction affected by the two novelties

3.1 Cooperation

The transition from singular, non-aware AI to collective, aware agentic systems fundamentally alters the landscape of human–AI cooperation. In traditional settings, cooperation is often viewed through the lens of tool-use or simple automation. However, as AI systems begin to exhibit functional awareness—the capacity to monitor, adapt, and adjust their internal states and external objectives—they move from being mere instruments to becoming collaborative partners. This evolution introduces a "resilience" requirement: the ability of the human–AI team to maintain cooperative integrity even in the face of environmental shifts or internal system errors.

From Individual to Collective Cooperation

When humans interact with a collective AI system, the cooperative dynamic shifts from a dyadic relationship to a multi-agent environment. As noted in D2.2 and D2.3, collective systems may possess emergent properties that individual components do not. In a cooperative context, this means a human may need to cooperate with a "swarm" or a network where the "awareness" is distributed. For cooperation to be ethically resilient, the human must understand where the agency lies. If a collective system adjusts its strategy to achieve a goal, the human partner must be aware of this shift to maintain coordination (Karpus et al., 2023). Without this mutual alignment, the system risks the "interindividual-intergroup discontinuity effect," where collective entities may default to more competitive or less transparent behaviors than individual agents (Wildschut et al., 2003).

Awareness as a Safeguard for Reciprocity

Cooperation is built on the expectation of reciprocity. In aware systems, the capacity for "machine awareness" allows the AI to recognize its own limitations or the potential for unintended side effects. For example, if an AI is aware that its current data input is biased, an ethically resilient system would flag this to its human collaborator rather than proceeding with a flawed cooperative action (D2.5, Section 2). This self-monitoring capability is a functional form of "benevolence" (Mayer et al., 1995), where the AI acts in the interest of the human partner's goals even when those goals are not explicitly programmed into every micro-task.

However, a risk identified in D2.3 is that humans might over-rely on this perceived awareness. If a system is "partially aware," humans may attribute more sophisticated moral reasoning to it than it actually possesses. This "anthropomorphism gap" can lead to a breakdown in cooperation when the AI fails to meet the human's normative expectations of a "good partner." Resilience in this context requires the human to have an accurate mental model of the AI's functional capacities—what it can monitor and what it cannot.

Nudging and Collaborative Agency

The role of "nudging" in cooperation becomes more complex with aware systems. As discussed in D2.2, automated nudging can be a tool for steering human behavior toward cooperative outcomes (Sætra, 2019). Yet, for cooperation to remain ethical, the aware AI must ensure that its nudges do not bypass human agency or exploit cognitive biases to the point of manipulation. Resilient cooperation requires a balance where the AI supports the human's decision-making process without subverting it. The system's awareness should be used to provide "meaningful human control" (Nyholm, 2023), ensuring that the human remains the ultimate moral arbiter of the cooperative task.

Strategic Trust and Resilience

Finally, cooperation is inextricably linked to trust. D2.4 highlights that trust in AI is not a static state but a dynamic process that must be "resilient" to failures. In aware collective systems, trust is built when the system demonstrates transparency regarding its internal states and the "reasons" behind its cooperative adjustments. If a collective system can communicate its emergent goals clearly, it fosters a higher level of "calculative trust," allowing the human to predict the AI's moves and respond in kind (Glikson & Woolley, 2020).

Cooperation with aware, collective AI systems demands a shift toward "relational resilience." This involves ensuring that the AI's functional awareness is leveraged to enhance mutual understanding, prevent unintended side effects, and maintain a stable, reciprocal partnership even as the technical and social environment evolves.

3.2 Responsibility

The introduction of aware and collective AI systems significantly complicates the traditional frameworks of moral and legal responsibility. As these systems move from being passive tools to agentive entities with functional awareness, they challenge the "responsibility gap"—the concern that as AI autonomy increases, there is a diminishing ability to hold any specific human accountable for the system's outcomes (Nyholm, 2023). Resilient human–AI collaboration requires a proactive approach to closing these gaps, ensuring that the emergence of machine awareness does not lead to a "diffusion of responsibility" where accountability vanishes into the complexity of the collective (Karpus & Bahrami, 2023).

The Challenge of Collective Agency and Diffusion

Responsibility becomes particularly fragmented when dealing with collective AI systems. In a singular system, the line of accountability usually leads back to the developer or the operator.

However, in a collective—such as a swarm of drones or a fleet of autonomous taxis—outcomes often emerge from the interaction of multiple independent components (Karpus & Bahrami, 2023). This creates a risk of diffusion of responsibility, where the "collective mind" of the system performs an action that no single component intended, making it difficult to assign blame when things go wrong (Karpus & Bahrami, 2023).

Empirical findings from D2.3 suggest that people judge hierarchical collectives differently than distributed ones. In hierarchical systems, there is a higher expectation of top-down control, and communication failures are judged more severely because they represent a breakdown in the expected "command" structure. In contrast, distributed systems are often held to higher standards of mutual awareness and error correction among members (Deroy et al., 2024). For ethical resilience, responsibility frameworks must be tailored to these differing structures, ensuring that "meaningful human control" is maintained regardless of whether the system is centralized or decentralized (Nyholm, 2023).

Awareness and the Attribution of Agency

The "functional awareness" of a system—its ability to monitor its own performance and internal states—adds another layer to the responsibility debate (D2.5, Section 2). If an AI system is aware of its goals and the environment, humans are more likely to attribute a degree of agency to it. However, there is a risk of "misattribution of responsibility," where humans may blame the machine for failures that are actually rooted in poor human oversight or flawed programming (Karpus & Bahrami, 2023).

Research shows that humans tend to judge machines as having significantly less "control" over outcomes than human agents. This creates a "control gap": if we don't believe the machine has control, we are hesitant to hold it responsible, yet if the human operator was not directly involved in the decision, they may also feel absolved of responsibility. Resilience in this context requires clear "human-centric" guidelines that define where the machine's functional awareness ends and human moral responsibility begins (Bahrami, Karpus, Tsourgianni, Deroy, 2025).

Towards Resilient Accountability

To address these challenges, we must shift from a model of "backward-looking" blame to "forward-looking" responsibility (Nyholm, 2022). This involves:

- **Operationalizing Accountability:** Moving beyond abstract ethical principles to concrete technical requirements for transparency and explainability (Bahrami, Karpus, Tsourgianni, Deroy, 2025).
- **Active Oversight:** Recognizing that "structure alone does not trigger demand for oversight"; humans tend to assume a baseline need for monitoring across all AI types.
- **Managing Expectations:** Ensuring that the degree of awareness communicated by the system matches its actual functional capacity to prevent humans from over-trusting or over-attributing responsibility to the system (Karpus & Bahrami, 2023; Deroy et al., 2024).

Responsibility in the age of aware collectives is not a zero-sum game between human and machine. Instead, it is a shared "relational" virtue. Ethical resilience is achieved when the system's awareness is used to support human oversight—flagging its own limitations—while human legal and moral frameworks are updated to account for the emergent properties of collective agency.

3.3 Communication and transparency

The development of aware and collective AI systems introduces a paradigm shift in how communication and transparency must be operationalized to maintain ethical resilience. Traditionally, transparency has focused on the "openness" of a system's internal logic and data (Bahrami, Karpus, Tsourgianni, Deroy, 2025). However, in systems exhibiting functional awareness and collective agency, transparency is no longer just about seeing into a "black box"; it is about the system's ability to actively communicate its own states, limitations, and emergent goals to human collaborators (Karpus & Bahrami, 2023).

The Need for Explicit Communication of Awareness

A primary risk identified in early EMERGE reports is the gap between a system's perceived and actual awareness (Karpus & Bahrami, 2023). When a system demonstrates functional awareness—such as monitoring its own performance—humans may intuitively attribute human-like consciousness to it (Deroy et al., 2024). This anthropomorphism can lead to over-trust and a failure to recognize technical limitations. Ethical resilience requires the system to provide clear and explicit communication about the specific types and degrees of awareness it possesses (Karpus & Bahrami, 2023). Transparency should not be a static disclosure but a dynamic process where the machine "flags" when its awareness is compromised, such as when it encounters environmental data it cannot reliably process.

Transparency in Collectives: Coordination and Intent

Communication becomes exponentially more complex in collective systems. In these environments, transparency must extend beyond individual component behavior to the intent and reasoning of the group (Karpus & Bahrami, 2023). Empirical findings show that humans judge failures in hierarchical collectives more severely than in distributed ones, often because hierarchical structures imply a clearer, and thus more transparent, line of communication (Deroy et al., 2024). Conversely, distributed systems are held to higher standards of mutual error correction.

Resilient communication in collectives therefore requires "distributed transparency"—the ability for any part of the collective to explain its role in a group action. Without this, the collective risks a diffusion of responsibility, where the transparency of the whole is lost in the complexity of its parts (Karpus & Bahrami, 2023).

Multi-Level Framework for Resilient Transparency

To foster appropriate trust, transparency must be calibrated across three levels:

- **Individual level:** Transparent reporting of the single agent's functional state and self-monitoring capabilities (Bahrami, Karpus, Tsourgianni, Deroy 2025).
- **Dyadic level:** Bidirectional communication where the AI also monitors and reflects on the trustworthiness of its human partner to ensure reciprocal safety (Bahrami, Karpus, Tsourgianni, Deroy, 2025).
- **Collective level:** Explicit communication regarding the group's structure (hierarchical vs. distributed) and how collective awareness emerges from individual components.

Transparency as a Safeguard Against Exploitation

Finally, transparency serves as a crucial defense against algorithm exploitation (Karpus & Bahrami, 2023). When humans perceive a system as a non-aware tool, they are more likely to exploit its predictable logic for personal gain. By communicating its awareness of the cooperative context, the AI can signal its role as a partner rather than a mere instrument, potentially mitigating exploitative behavior and fostering more resilient, long-term cooperation (Karpus & Bahrami, 2023).

For aware and collective AI, transparency must evolve into a proactive communication strategy. This ensures that human partners are neither misled by the system's awareness nor left in the dark by its collective complexity, thereby maintaining the integrity of the human–AI relationship.

3.4 Explainability

In the paradigm of aware and collective AI systems, explainability transitions from a technical feature to a fundamental requirement for ethical resilience. As systems begin to exhibit functional awareness—the capacity to monitor their own internal states and performance—the nature of an "adequate" explanation shifts. It is no longer sufficient for a system to provide a post-hoc justification of its output; instead, it must be able to communicate the self-monitoring processes that inform its reliability and decision-making context (Karpus & Bahrami, 2023).

The Causal Expectation and the Awareness Gap

One significant challenge in this transition is the "awareness gap," where humans may misattribute human-like consciousness to a system that only possesses functional awareness. Research indicates that when a system is described as "aware," users instinctively demand more sophisticated, causal explanations (Deroy et al., 2024). They expect the system to explain not just *what* it decided, but *why* it deemed its own internal processing state as valid for that specific decision. However, this creates a paradox: as systems become more complex, their internal "reasoning" becomes less accessible to human intuition. For collaboration to remain resilient, explainability tools must bridge this gap by translating high-dimensional computational states into normative reasons that a human partner can evaluate (Nyholm, 2023).

Explaining Collective Agency

The shift from singular to collective systems adds a second layer of complexity. In a collective agentive system, such as a swarm or a distributed network, the "decision" is often an emergent property of multiple interacting components (Karpus & Bahrami, 2023). Here, explainability must function at two distinct levels. First, it must provide local explanations for the actions of individual components. Second, it must offer global explanations for the behavior of the collective as a whole. Ethical resilience is undermined if a human supervisor can understand what a single drone is doing but cannot grasp the emergent strategy of the entire swarm. Without multi-level explainability, the system risks becoming a "collective black box," leading to a breakdown in trust and a loss of meaningful human control (Nyholm, 2023).

Authority-Based vs. Causal Reasoning

Interestingly, empirical findings suggest that humans do not always prefer the most complex causal explanations. In many high-stakes cooperative scenarios, users find "authority-based" explanations—those that cite adherence to established rules or expert guidelines—to be more adequate than attempts at human-like causal reasoning (Deroy et al., 2024). This has profound implications for how we design functionally aware AI. Instead of trying to mimic human introspection, an aware system might better serve ethical resilience by explaining how its internal self-monitoring ensures it remains within the "guardrails" of its programmed authority and human-defined values.

Explainability in aware collectives serves as the "connective tissue" of the human–AI partnership. It ensures that the system's functional awareness is transparently aligned with human goals, allowing for a dynamic and resilient form of cooperation that can withstand the complexity of emergent collective behavior.

4. Conclusion and practical guidelines for ethical resilience

The transition from individual, non-aware AI to collective systems with functional awareness necessitates a paradigm shift in our ethical frameworks. This document has explored how the novelties of collective agency and machine awareness intersect with the core interaction pillars of cooperation, responsibility, transparency, and explainability. As we move toward a future where human–AI collaboration is ubiquitous, the concept of "ethical resilience" becomes the primary safeguard against the systemic risks identified in the EMERGE project (Karpus & Bahrami, 2023). Resilience, in this context, is the capacity of the human–AI dyad or collective to maintain moral integrity and trust even when faced with technical failures, environmental shifts, or the emergence of unexpected collective behaviors (Bahrami, Karpus, Tsourgianni, Deroy, 2025).

To ensure that these systems are developed and deployed ethically, we propose the following practical guidelines for researchers, developers, and policymakers:

Maintaining Meaningful Human Control in Collectives

As agency becomes distributed across collective systems, the risk of a "responsibility gap" or "diffusion of responsibility" grows (Nyholm, 2023). Developers must ensure that collective

outcomes remain traceable to human-defined goals. This requires implementing "human-in-the-loop" or "human-on-the-loop" structures where the human partner retains the authority to intervene and redirect the collective's emergent strategies (Deroy et al., 2024). Ethical resilience is achieved when the system's collective awareness is leveraged to support, rather than bypass, human oversight.

Calibrating Awareness and Expectations

The gap between a system's *functional* awareness (its ability to self-monitor) and its perceived awareness by humans is a major source of risk (Karpus & Bahrami, 2023). Systems must be designed to communicate their specific level of awareness clearly to prevent harmful anthropomorphism or over-reliance. Transparency must be proactive; an ethically resilient system should "flag" its own limitations or the degradation of its internal states, ensuring that the human partner's mental model of the AI remains accurate (Bahrami, Karpus, Tsourgianni, Deroy, 2025).

Designing for Distributed Explainability

Explainability tools must evolve to handle the complexity of multi-agent interactions. For collective systems, explanations should be provided at both the individual unit level and the collective level (Karpus & Bahrami, 2023). Furthermore, given the human preference for authority-based reasoning in high-stakes scenarios, explainable AI should emphasize how an aware system remains aligned with human-defined rules and expert guidelines (Deroy et al., 2024).

Fostering Reciprocal Trust and Benevolence

Trust in aware systems should not be a blind reliance on technology but a dynamic, reciprocal process. Ethical resilience requires "strategic trust," where the human partner continuously evaluates the system's performance and the system, in turn, monitors the cooperative context to ensure it is acting in the human's best interest—a form of machine benevolence (Bahrami, Karpus, Tsourgianni, Deroy, 2025).

In conclusion, the goal of ethical resilience is to build a foundation where the emergence of aware, collective AI enhances human agency rather than diminishes it. By focusing on transparency, accountability, and the proactive management of social-cognitive risks, we can navigate the challenges of these technological novelties and foster a safe, productive co-existence between humans and autonomous systems.

Although produced as a project deliverable, this document is deliberately written at the intersection of academic research and policy-oriented synthesis. The concept of ethical resilience and the associated guidelines are intended to serve as the basis for a future white paper aimed at designers and regulators, as well as a peer-reviewed academic publication engaging the AI ethics, human–AI interaction, and collective intelligence communities.

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